

# **GAME DESIGN DOCUMENT**

# **PROJECT IOTA**

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## Game Analysis

The basic gameplay loop consists of the player being roused from cyrosleep periodically, allowing them to complete their tasks. The player is time limited to complete these tasks due to their own limited suit energy, and their suit's heat management system. The main challenge revolves around the player spending their limited time as efficiently as possible to keep the habitat in one piece for the duration of their next cryosleep. Nearing the end of the game, the player will be presented with a moral dilemma that requires them to decide whether to maintain the station's recently discovered malicious intentions, or allow the station to be destroyed, killing themselves and all their colleagues on board.

## Mission Statement

Project Iota is a story based, atmospheric game with interaction and management-based gameplay elements. The player takes the role of a station maintenance engineer that is tasked with keeping the atmospheric habitat from being destroyed by managing the Hull Integrity, the Heat, the Oxygen Levels, and the Weight of the habitat. If the player fails to successfully keep these aspects of the habitat in check, the habitat will be destroyed, killing the player and all the scientists on board.

## Flavor Blurb

Welcome to Project IOTA, Engineer! You have been chosen and trained to maintain the state-of-the-art Inverse Orbital Trajectory Apparatus. Floating at neutral buoyancy within the atmosphere of gas giant 6D Alpha Infernus, the station requires regular maintenance to due extreme temperatures, limited oxygen, and the balloon-based buoyancy system. Careful management of time is required due to your suits limited capability to combat the extreme heat of the station, as engineer are not rated for long-term consciousness onboard the IOTA. We have every faith you will succeed and are looking forward to your service.

## Genre

Project IOTA is a narrative driven, first-person atmospheric game with light survival mechanics.

## Platforms

PC, Unity Game Engine.

## Target Audience

Project IOTA is aimed towards players who enjoy story rich scenario games similar to *Stanley Parable* and *Firewatch*. The moral dilemma explored in the game's narrative may be found unsuitable for younger audiences.

## Storyline & Characters

Project IOTA is a government funded science excursion that takes place in the state-of-the-art Inverse Orbital Trajectory Apparatus, a long-term, interstellar habitat floating in the atmosphere of the volatile gas giant 6D Alpha Infernus. The player is a recently hired engineer whose task is to maintain the station regularly, as the unstable atmosphere of the planet is constantly degrading the integrity of the heat shielding, disabling life support systems, and affecting the buoyancy of the IOTA's balloon system. Without the player's constant maintenance, the station will reach a critical state and become destroyed or otherwise non habitable to themselves and the scientists on board.

Gas Giant 6D Alpha Infernus has a fluctuating weather cycle and a very slow rotation speed, resulting in periods of research safe "dark cycle" periods where the geostationary orbit of the IOTA facility is shielded from the local star's intense heat and radiation, and unsafe "light cycle" periods where the facility is exposed to direct starlight of the close proximity star, creating a non-survivable living condition where employees must enter cryosleep until the next dark cycle. The Engineer is responsible for performing ongoing maintenance to the IOTA facility during these dangerous light cycles to ensure the survival of themselves, the project, and their colleagues aboard the station.

Unbeknownst to the player, their contribution to maintaining the station is furthering a more sinister agenda. During one of their routine maintenance light cycles, the player learns information that creates an air of question as to whether they should continue to participate in Project IOTA, but they cannot simply abandon their colleagues to die either. This will give the player a moral choice in the story and gameplay elements, giving more reasoning and weight to the choices they make towards the end of the game.

| Character                                                                                                   | Description                                                                                                                                          | Characteristics                                                                                                                             | Misc. Info                                                                                                   |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <div>The Engineer</div>  | The player character. The engineer is hired onto Project IOTA to maintain the station, and ideally not poke around anywhere they aren't supposed to. | The engineer must maintain the station's oxygen, heat, and buoyancy. Their suit is equipped to assist with this, but it has limited energy. | The player will gain access to information later in the game that presents player choice in a moral dilemma. |

# Gameplay

## Overview of Gameplay

Project IOTA features a rich atmosphere and compelling narrative to drive the player. This method of environment and writing based story telling can be compared to games like *Stanley Parable* or *Firewatch* where the player moves to locations in the game in an exploration driven storyboard that offers choice but is not completely open ended. The game will also include light survival elements to provide urgency and meaningful decision making in gameplay. The decisions the player makes are designed to result in multiple endings based on how the player chooses to act in the later stages of the game. The game will only be single player and will not feature any online functionality.

## Player Experience

You are an Engineer for the IOTA Science Facility suspended in neutral buoyancy in the atmosphere of gas giant 6D Alpha Infernus. Your job at this facility is to perform routine maintenance during the dangerous Light Cycles where the facility is exposed to the direct starlight and radiation of the nearby star. Due to the proximity of the planet to the star, the station's temperature is extremely high, requiring ongoing heatshield repair, buoyancy checkups, and life support maintenance to keep yourself, your colleagues, and the project alive. To combat the life-threatening effects of the starlight exposure at the facility, employees are instructed to remain in cryogenic sleep pods until the next Dark Cycle allows for safe research conditions. You are expected to leave your cryopod during Light Cycles to perform scheduled maintenance, ensuring the health of the station and the crew. You have not been given the security clearance to access any of the scientist's data pads, and you are not cleared to have the goal of project IOTA disclosed to you, however, you are the only one conscious on the station during Light Cycles and there isn't anyone to stop you. You will be tasked with managing the station's needs every maintenance period, but you must be wary of your exposure to the starlight. Your suit is only rated for short bursts of heat and radiation before it must return to its respective cryochamber for decontamination and cooldown, which is a lengthy process. This limitation will require you to complete your tasks efficiently and choose how you spend your time wisely. The fate of your colleagues and project IOTA is in your hands.

## Gameplay Guidelines

- The player's time is limited by their exposure to heat during their active play periods.
- The player is expected to meet a minimum maintenance quota to succeed at their objective for each Light Cycle, otherwise the game ends.
- The player is given the choice to spend some of their time reading up on Project IOTA, but this must be juggled on top of their required maintenance tasks
- With the knowledge the player gains from the information scattered throughout the station, the player can decide the fate of the entire project.
- Heat is the main driving factor for the player's urgency. The player cannot let their suit overheat or else the game will end.

## Game Objectives & Rewards

The game objective for the player is to successfully maintain the station in addition to learning the mystery behind project IOTA. Since the player's status as Engineer disallows them from knowing the motives behind the project, they must take it upon themselves to learn about what exactly is going on, even if it isn't explicitly allowed. The reward for continuous maintenance means that the player is allowed to continue onto the next Light Cycle, allowing them to learn more about the facility's goals as their colleagues' research continues. This eventually culminates to the player facing a moral dilemma where they are given a choice on how to handle their situation, generally resulting in the fate of themselves and the station.

## Gameplay Mechanics

- **Heat:** Heat is the player's main "rival mechanic" where it dictates how much they can accomplish at a given time. The player's suit can only withstand so much exposure to the starlight and radiation that impacts the station, so the player cannot stray too long from their objective.
- **Life Support:** Life support is one of the maintenance locations for the player to manage. Ideally, this location is nearby the cryogenic sleep chambers, allowing the player to choose to work on this part of the station immediately after waking, or just before sleeping again. The temperature of this location will be low, allowing the player to spend more time here.
- **Buoyancy:** Buoyancy of the facility is managed in the furnace control room where the burner of the station is located. Here, a special type of controlled nuclear fusion takes place, allowing the station to stay afloat. The player must increase or decrease the intensity of the burner to have the station ascend or descend slightly. The heat in this location is moderate.

## Gameplay Mechanics (Continued)

- **Shielding:** The shielding of the station is located along the outside scaffolding of the station, directly exposed to the starlight and the elements of the atmosphere of the planet. The player must repair certain segments of the shielding to maintain the station. The heat in this location is the most intense. The player must be sure to only spend as much time here as required.
- **Datapads:** Datapads are informational tidbits from the station's scientists that provide information on what the motives for Project IOTA are. Since the player is not given the information directly about what is going on at the station, they are able to learn about the situation from the datapads scattered around the station. As they wake into the later Light Cycles, more information becomes available to the player. Some datapads with more critical information regarding the malicious intent of the project require more time for the player to reach, requiring the player to plan around how they use their time.

## Level Design

The game's level design will focus on a singular level. This level must have a layout that requires the player to think about committing to a route due to the time investment of each task in addition to the travel time required to arrive there. It will be important to balance this design obstacle in a way that provokes thought and choice without making the layout confusing or a chore to traverse. Each maintenance local can feature a "time sink" that the player must commit to if we find that the player is finishing their objectives too quickly. An example of this mechanic would be a large radiation shielding door that the player must spend time opening and closing when working in the burner room. To avoid additional commitments to adding this system, the mechanic can simply feature a simple "Hold to Use" system without the need of complex animations or programming.

Datapads containing information for Project IOTA's intentions can be found every light cycle. Datapads featuring more nefarious information are available in later Light Cycles and in locations that require more time investment to read. This provides a situation where the player must make a choice on how they will spend their time each cycle before they must return to their cryogenic sleep chamber.

Each section of the level will feature a varying level of heat that is either higher or lower than normal. This will also influence how the player decides to move around the station and how to prioritize their tasks and extra time for exploration.

## Control Scheme

The player will be controlled using a first-person movement system. The only platform planned for the game currently is Unity PC, so the character will be controlled via WASD movement keys. Additionally, the player will be able to interact with some objects using a key, which will be defaulted to E.

## Game Aesthetics & User Interface





## HUD Ideas

